5 Additional objects and experimental design

The square of Section 4 stretches almost uniformly and has a known answer. Real parts have holes, corners and slender members. This raises three questions: how shape changes the deformation, whether it changes A and hence the cost of solving, and whether results from a regular grid carry over to an irregular mesh. We set up four objects and a list of comparisons, each changing one factor at a time; Section 6 reports the outcomes.

Figure 1 shows the objects, with lengths in metres. The square S is the plate of Section 4 on an irregular mesh. The plate H adds a central hole of radius 0.15; in a wide plate under tension, the stress at a small hole is three times the applied stress [3]. The bracket L has counterclockwise vertices $(0,0)$, $(1,0)$, $(1,0.5)$, $(0.5,0.5)$, $(0.5,1)$, $(0,1)$. At its inner corner the exact stress is infinite [4], so stress there cannot settle under refinement. The beam B, $0 \le x \le 4$, $0 \le y \le 0.5$, bends, which linear triangles represent poorly on coarse meshes [1].

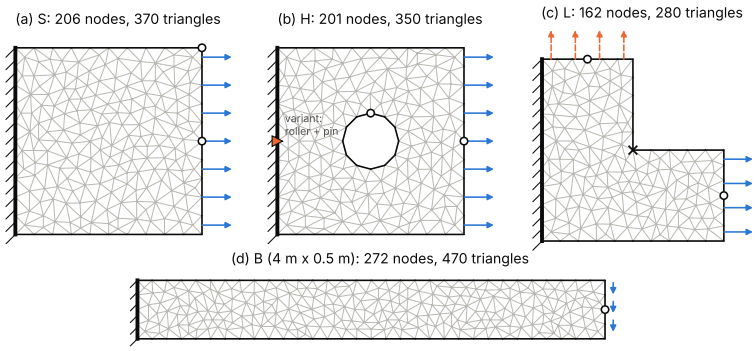


Figure 1: The four objects on their coarsest meshes. Hatching: clamped edge. Blue arrows: baseline load. Dashed orange arrows: alternative load on L. Orange triangle: pin used with a roller support on H. Circles: points where displacements are compared; cross: inner corner of L.

We mesh each object with Triangle [2]. Its triangles follow the boundary, leave the hole empty and have no angle below 30°. Four levels, with mesh size h from 0.1 to 0.025 (beam: 0.125 to 0.031), give $N \approx 300$ to 7800 unknowns. The comparison points are nodes on every level.

Material and thickness are those of Section 4. Every object is clamped on $x = 0$, which stops both translations and the rotation, so A is symmetric positive definite; we confirm that its smallest eigenvalue is positive. S, H and L carry the traction $p = 100$ MPa on their face $x = 1$. The beam carries a downward tip traction of 10 MPa, a force $P = 50$ kN; beam theory predicts a tip deflection $PL_B^3/(3EI) = 51.2$ mm, with length $L_B = 4$, $I = tW_B^3/12$ and depth $W_B = 0.5$ [3]. Loads act on the same boundary pieces on every mesh, so each level carries the same total force.

Table 1 lists the comparisons. Load size, load position and E change only b or scale A, so their effect is known in advance and tests the code; ν and the supports change A itself. Under refinement, changes at the comparison points estimate the discretization error, while κ_2 and iteration counts show how solver cost grows with N.

Every system is solved by sparse LU, CG, diagonal PCG, MINRES and Gauss–Seidel under the rules of Section 3, and checked as in Section 4.

References

[1] Robert D. Cook, David S. Malkus, Michael E. Plesha, and Robert J. Witt, *Concepts and applications of finite element analysis*, 4th ed., Wiley, New York, 2002.

Table 1: Case register. Variants of H, L and B use the second mesh level. u_P: displacement at a comparison point; iter.: iteration counts.

Study	Question	Changed	Fixed	Measured
Geometry	effect of shape	S, H, L	load, clamp, h	u_P, κ_2, iter.
Refinement	convergence, cost	h, 4 levels	object, load	u_P, stress, κ_2, iter.
Mesh type	grid vs irregular	S vs grid, $m = 10$–40	square problem	u_P, κ_2, iter.
Load size	linearity	$p = 25$, 100, 250 MPa (H)	A	u_P/p, iter.
Load location	position	face $x = 1$ vs $y = 1$ (L)	A, total force	u_P, iter.
Supports	clamp vs roller	roller on $x = 0$ + pin (H)	load, mesh	u_P, κ_2, iter.
Material	role of E, ν	$\nu = 0$, 0.3, 0.45; $E = 70$ GPa (H)	shape, load	u_P, κ_2, iter.
Bending	beam accuracy	h, 4 levels (B)	beam, P	tip u_y vs theory, κ_2
Load type	axial vs bending	tip load direction (B)	A	tip u, iter.

[2] Jonathan Richard Shewchuk, *Triangle: Generating Delaunay triangulations*, Carnegie Mellon University, Triangle documentation, `https://www.cs.cmu.edu/~quake/triangle.delaunay.html`. Accessed 10 November 2026.

[3] Stephen P. Timoshenko and J. N. Goodier, *Theory of elasticity*, 3rd ed., McGraw-Hill, New York, 1970.

[4] M. L. Williams, *Stress singularities resulting from various boundary conditions in angular corners of plates in extension*, Journal of Applied Mechanics **19** (1952), no. 4, 526–528.